Object: Shielded Enemy

Normal

Defeated

Properties	• Facing direction is always oriented to the player • The front part is always protected while the rest is vulnerable • Shield only covers the very front of the enemy
Behaviours	• Always rotating to face the direction of the player • Can't block attacks that are hit from its side or behind it
Game State Feedback	**Visual** • Enemy shuffles feet when rotating towards player • Enemy shrugs shield and shoulders when moving • Eyes glow when the enemy successfully blocks an attack **Audio** • Shuffling sounds when the enemy rotates • Small grunt when the enemy blocks an attack and moves
Systemic Feedback	The enemy's defense depends on the location of the player

Object: Shield

Properties	• Completely invulnerable • Only effective from it's front
Behaviours	• Rotates alongside its assigned Shielded Enemy • Can block any attack
Game State Feedback	**Visual** • Sparks fly off shield when it is hit by something • Small glow when enemy/player stops moving • Shakes a little when enemy moves it **Audio** • Scraping sound when enemy is rotating • Ominous sound effect when enemy stops rotating • Clang sound whenever something hits it
Systemic Feedback	Capable of protecting anything behind it from any type of attack

Object: Fire Staff

Properties	• Grants the skill 'Fire Ball' • Can be picked up • Item needed to solve the next few puzzles
Behaviours	• Floats in the world till player picks it up • Fires a damaging 'Fire Ball' • When equipped, it appears in the player's hand
Game State Feedback	**Visual** • Small glow in its tip when item isn't picked up • Stronger fiery glow when item is equipped • Embers sprout out when the Fire Staff is used • The staff sways slightly when the player moves while its in hand **Audio** • Click sound when the player picks up, and/or equips it
Systemic Feedback	Equipping this item grants the player a new usable mechanic

Object: Fire Ball

Player Fire Ball

Stone Dragon Fire Ball

Properties	• A fire type attack • Deals damage • The player can only use this when equipped with a 'Fire Staff' • Stone Dragons fire it as well
Behaviours	• Moves at a slow pace • travels in a direct line • Damages enemies • Deflects other 'Fire balls' • When two fire balls deflect each other, they both go in the opposite direction, to the left or right • Can activate/deactivate levers
Game State Feedback	**Visual** • Glows orange when its shot • Leaves a trail of embers • After hitting a surface or object, leaves a small burned spot • Has a slight explosion on impact with anything **Audio** • Fire sound when it is shot • Explosion sound effect on impact
Systemic Feedback	Timing the shot of the fireball can be used to solve different situations like damaging enemies, activating levers or deflecting other fire balls and potentially causing a chain reaction.

Object: Stone Dragon

Normal

Defeated

Properties	• Can be destroyed • Made of stone • Appears on walls
Behaviours	• Shoots 'Fire Balls' in a certain direction periodically • The player can destroy it using their own 'Fire Ball'
Game State Feedback	**Visual** • Opens its mouth when it shoots a 'Fire Ball' • Eyes glow red as well when it shoots • Slight smoke trail after firing a 'Fire Ball' **Audio** • Breath sound after shooting its projectile • Clamp sound when it closes its mouth • Occasional growl
Systemic Feedback	Acts as a way to force the player to be weary of their surroundings due to the constant projectiles the Stone Dragon is firing

Object: Lever

Deactivated

Activated

Properties	• Interactable • Unlocks something
Behaviours	• The player can activate and deactivate it using a fire ball
Game State Feedback	**Visual** • Upon being interacted with, the lever visibly moves/switches • Small sparks will appear when it moves **Audio** • Metal scraping sound when it moves • When it is finished its movement, a high or deep, beep will be audible, dependent if it is being activated or deactivated
Systemic Feedback	The lever is used to activate/unlock something in the vicinity

Scenario #1

A

1. The Player enters the room
2. The Shielded Enemy faces towards player
3. Player grabs Fire Staff

B

1. Player shoots a Fire Ball at the Shielded Enemy
2. The Shielded Enemy blocks the attack
3. The Player moves
4. The Shielded Enemy faces the Player

C

1. Player shoots a Fire Ball
2. Then they quickly move to the side
3. The Shielded Enemy moves to face the player
4. The Fire Ball hits and kills the Shielded Enemy, freeing up the exit door

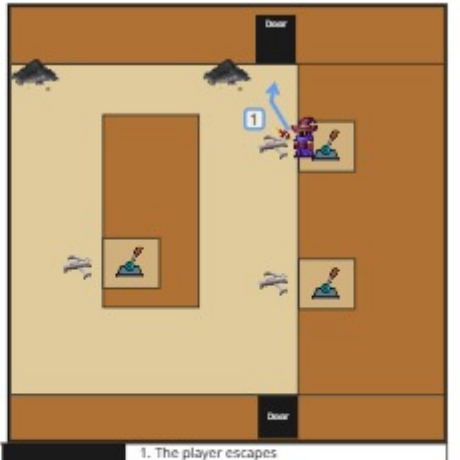
Scenario #2

A

1. The Player enters the room
2. The Player uses their fire ball to deflect the Stone Dragon's, defeating a Shielded Enemy
3. They deflect it again to activate the lever
4. Then moves and does the same thing, until they reach the top/back of the room

B

1. Then the Player destroys both Stone Dragons
2. The player destroys the right side enemy using the method from scenario #1
3. They go and activate the last lever



C

1. The player escapes

Assignment #2: Systemic Games with Emergent Mechanics

Course: GAME 10017 Game Systems Design

Professor: Raphaël Tétreault

Designer: Michael Gouveia

Date: 2026/02/03

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Mechanic(s) Justification

Fire Staff

The 'Fire Staff' acts as a way to give the player a sense of progression within the game, and locking a powerful ability that can only be used once its picked up. It is a separate object that opens up new interactions within the game world.

The weapon's mechanic is shooting a 'Fire Ball' in a straight direction when equipped by the player. It behaves quite simply, the player sees the staff, picks it up and can visibly see it in their hand when equipped, and it enables the player to cast damaging fire balls. It is an easy mechanic to get the grasp of, which allows more room for unique ways for the player to interact with other systems. It isn't only being used for raw damage, it creates opportunities when used against systems like the moving Shield Enemies, the Stone Dragons or Levers. It can be used to *deflect* other fireballs, causing both to go opposite directions to the left or right, and can possibly cause chain reactions.

There are clear visual, and audio, cues whilst using the Fire Staff; Whether it is from the staff appearing in the player's hand to distinct fire effects when casting a fire ball.

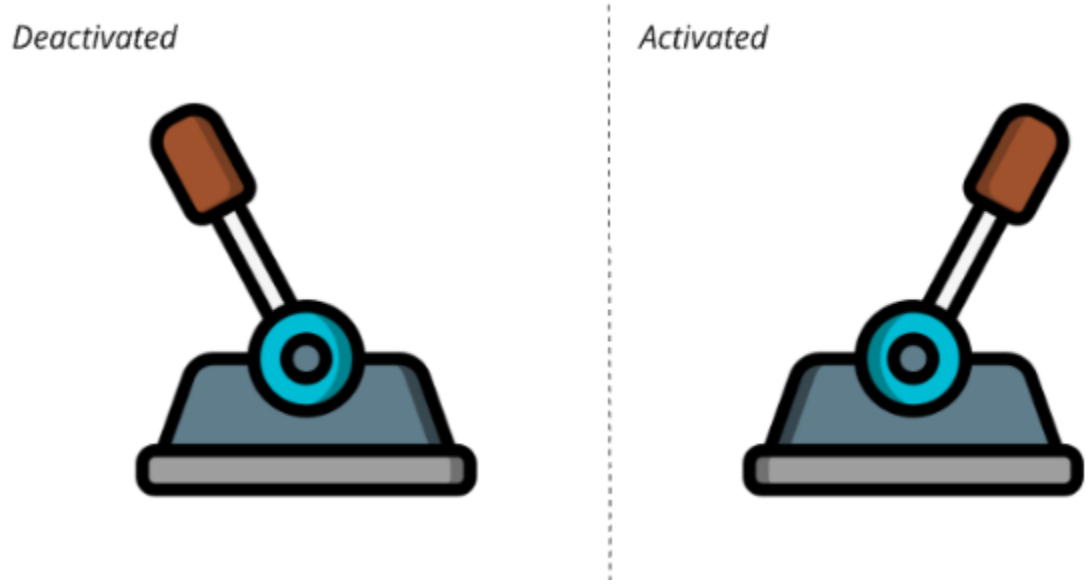
Shielded Enemy

This enemy is designed to rotate and always face the player, using its shield to protect itself from any attack towards its front. It encourages the player to use other systems like the Fire Staff and constant movement to expose its vulnerable parts. This is a very simple system and behaviour, which allows the player to quickly catch on how to deal with an enemy like this.

There is a lot of visual and audio feedback that further reinforces the Shield Enemy's system; The visual/sound of the rustling armor and shield when it rotates, grunts and eye glow, as well as feedback pertaining when the enemy successfully blocks an attack, and when it takes damage as well.

The Shielded Enemy offers the player a gameplay expirence focused on positioning and utilizing different mechanics/tools to be taken down, allowing the player to manipulate different elements and increase the possibilities within their given space.

Lever(s)



The lever is designed to be a simple inertactible object, that can be utilized to solve different puzzles; With only two different states, activated or deactivated, which can be used to unlock a door or deactivate a moving platform. The lever's behaviour is limited to this, which allows it to be connected to other systems within the game without any extra complexitities.

There are small notions of feedback that clearly shows what state each lever is in like a positive/negative beep sound. This system gives the player a simple cause and effect interaction that can be used to progress through different areas.

Stone Dragon

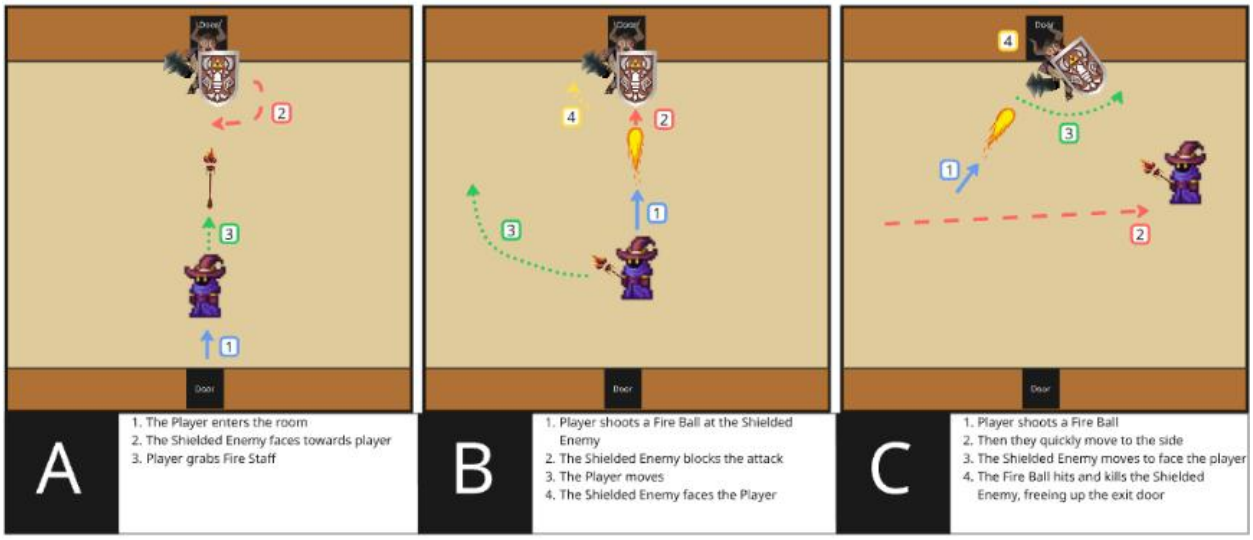
The Stone Dragon is an enemy/environmental threat that is designed to put constant pressure on the player. It's behaviour consists of firing fire balls in a certain direction, often shooting in paths to prevent the player from advancing, forcing the player to react instead of outwright attacking it.

The player can counteract the Stone Dragon's fire balls with their own, cancelling out incoming ones. This is a connection between two different systems, which allow the player to use what they know and tools they have to safely navigate the area.

There is clear visual and audio feedback that communicates to the player when the Stone Dragon is firing, destroyed or when its fire ball gets cancelled out by theirs. Systemically, the Stone Dragon turns the fire ball, which is usually used to cause damage, into a spatial control and timing challenge.

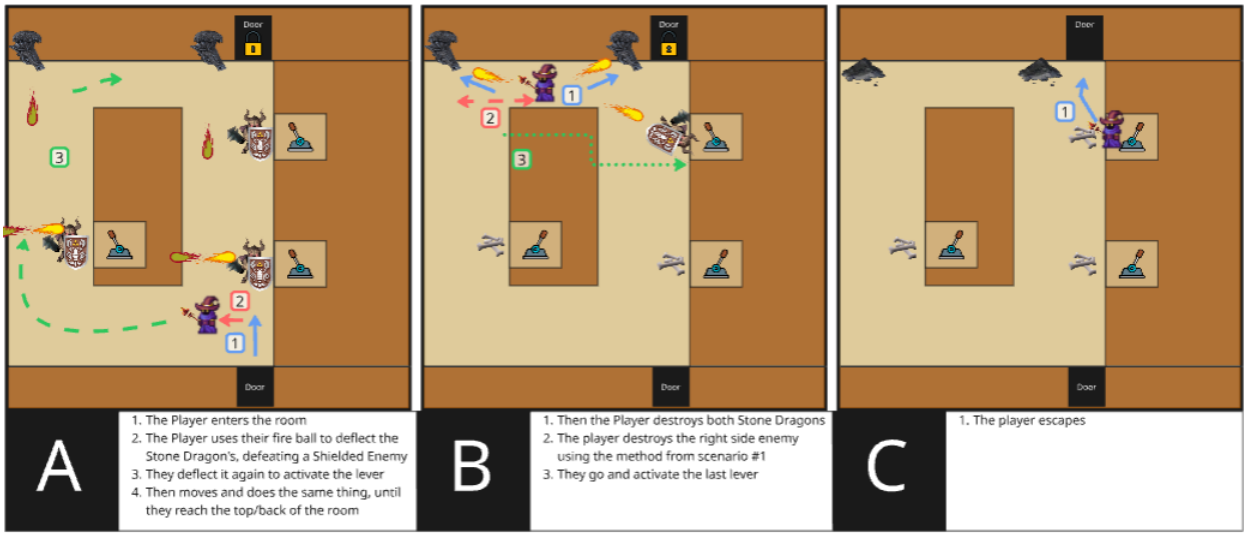
Emergence & Gameplay Space Possibilities Justification

Scenario #1



This scenario uses a minimal amount of objects and mechanics like the staff and fire ball, the shielded enemy, and the player, using directional and timing based systems. The shielded enemy's facing behaviour and its invulnerable, directional shield, creates multiple possibilities in terms of the player's movement and positioning. This allows this scenario to generate multiple solutions based on each objects properties.

Scenario #2



This scenario uses many of the core systems and properties from the first scenario, specifically the fire ball, to create a new gameplay scenario. The interaction between the player, the Stone Dragons, and their fire balls, generates a variation of different outcomes through projectile cancellations, postioning and movement, as well as timing. The objects in this scenario have flexible properties and systems that can be combined to allow for more possibilities within the space.